Thomas C. Fraser
tfraser@perimeterinstitute.ca
+1 (226) 868-0557
Perimeter Institute for Theoretical Physics,
31 Caroline St N, Waterloo, Ontario N2L 2Y5, Canada

January 15, 2023

Dr. Ludovico Lami,
Faculty of Physics, Mathematics & Computer Science,
QuSoft & University of Amsterdam.

Dear Dr. Ludovico Lami,

I am writing to express my interest in joining your research group as a post-doctoral student. I am currently a PhD Candidate working under the supervision of Rob W. Spekkens in the Quantum Foundations group at the Perimeter Institute for Theoretical Physics in Canada. I will be graduating spring 2023.

The quantum marginal problem is at the heart of many conceptual and technical challenges in quantum theory including the separability problem, the local-Hamiltonian problem, and multipartite entropic inequalities.

Recently, I found a complete solution to the quantum marginal problem for finite-dimensional local Hilbert spaces in the form of a convergent hierarchy of necessary operator inequalities (arXiv:2211.00685). This solution was the result of (i) conceptualizing the quantum marginal problem as an instance of asymmetric hypothesis testing, and (ii) deriving a leading order lower bound on the regularized max-relative entropy by improving some results from M. Keyl's work on quantum state estimation.

I have also worked on resource theories of entanglement (arXiv:2004.09194v2) and correlations in classical and quantum networks (arXiv:1709.06242v2, arXiv:1902.07091v2).

I am interested in this position because it would be a fantastic opportunity to develop and apply these methods further to topics including (i) (non-)asymptotic non-linear property testing, (ii) large deviation theory of quantum measurements, (iii) analytic bounds on relative entropies of resource, and (iv) the Gaussian & channel quantum marginal problems.

Thank you for your time and consideration.

Sincerely,

Thomas C. Fraser